

Gabriel Duarte Rengifo

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EDUCATION

Purdue University | 3.44 / 4.0

West Lafayette, IN

BS Aeronautical and Astronautical Engineering

Certificate in Entrepreneurship and Innovation

Aug 2025 — Expected May 2029

Pine Crest School

Fort Lauderdale, FL

High School Diploma

Aug 2022 — May 2025

WORK EXPERIENCE

Gulfstream Aerospace

May 2026 — July 2026

Additive Manufacturing Lab Test Intern

Savannah, GA

- Designed 7 custom tooling projects in CATIA applying DFM/DFA principles, owning each from concept through customer feedback to delivery; 2 were adopted into production on delivery, one clearing a 2-month work backlog and pulling a shipment roughly 1 month ahead of schedule
- Operated SLA, FDM, and SLS additive manufacturing machines (Formlabs, Bambu, Markforged, Stratasys, EOS) to produce shop aids and training tools for internal production
- Performed hands-on troubleshooting and preventive maintenance on AM equipment; redesigned lab floorplan to open dedicated space for composite layup work

Presidential Aviation

June 2024 — July 2024

Summer Intern

Fort Lauderdale, FL

- Audited and reconciled two years of maintenance records for 19 clients against FAA airworthiness requirements, identifying documentation errors across a 20+ aircraft fleet
- Compiled findings into a report for the maintenance department recommending consolidation of overlapping maintenance events to reduce aircraft downtime and automation of the manual record-entry steps producing the errors found

LEADERSHIP & TECHNICAL PROJECTS

Humanoid Robot Club

Aug 2025 — Present

President

West Lafayette, IN

- Leading a 1,500+ member organization across 3 technical subteams, setting club strategy, sponsor relationships, and technical project direction
- As President of StarkHacks, an HRC program, ran operations for a 36-hour in-person hardware hackathon with 750 attendees, the largest by attendance found in public records; owned venue contracting at the Purdue Armory, catering, AV/production, dedicated participant Wi-Fi infrastructure, and parking logistics
- Designed all 5 finger linkages, enclosure, and actuator/servo wiring for a fully articulated robotic hand; selected linkage actuation over cable-driven for grip strength and field reparability, delivering 3 CAD iterations and 2 assembled prototypes, with the current prototype verified grasping 2 lb payloads in ongoing testing
- Presented hand design research at Purdue's fall and spring undergraduate research conferences, listed under "Presentations with Distinction" at both

Purdue Space Program

Aug 2025 — Present

Active Controls Propulsion & High Power Rocketry Subteam Member

West Lafayette, IN

- Designed the actuator mounting scheme for a gimbaled thrust vector control system on PSP's Tadpole rocket; wrote a MATLAB tool evaluating 250+ candidate mount point sets against actuator stroke and max torque constraints using engine profile geometry extracted from NX CAD, narrowing to 3 configurations; the selected configuration is now in final design and fabrication
- Building a Level 1 and Level 2 high-power rocket (kit build, OpenRocket simulation, team mentorship) toward NAR/Tripoli certification as part of PSP's high power rocketry subteam

PlayAR

Oct 2023 — June 2026

Co-Founder

Fort Lauderdale, FL

- Co-founded PlayAR, an XR football training platform on Meta Quest Pro letting players run reps against virtual opponents in first-person; owned product direction and worked alongside the dev team on animation and movement UI, translating coach feedback into build requirements
- Piloted the prototype with our high school football team, running live play-throws against virtual players; coaches responded positively, with several continuing to integrate AR tools into practice since
- Advanced to the Diamond Challenge semifinal round as a top-600 team (2024) and to the Limitless Global Summit as a top-50 team (2025)

RELEVANT COURSEWORK

AAE 251: Intro to Aerospace Design

- Applied aerodynamics and orbital mechanics fundamentals via MATLAB problem sets and pitot tube analysis; collaborated in a 6-person team on conceptual design of a full aerospace system

MFET 163: Foundations in Geometric Modeling & Data Management

- Utilized NX and a PDM system (Teamcenter) to model parts, manage variants via expressions/design tables, and support an Engineering Change Process (ECOs) through a simulated industry PLM workflow

SKILLS

- Software:** Fusion 360, AutoCAD, SolidWorks, OnShape, Siemens NX, CATIA, MATLAB, Teamcenter, OpenRocket
- Technical Skills:** Additive Manufacturing, SLA/SLS/FDM, Computer Numerical Control (CNC), Computer Aided Design (CAD), Mechanical Design, Laser Cutting, DFM/DFA
- Programming Languages:** JavaScript/TypeScript, Python, Java, C
- Certifications:** FAA Private Pilot Certificate
- Languages:** English, Spanish